# Evaluation runs: **pbkv** (a2-store)

generated 2026-11-03 from brain-data 48587ddd0 2026-11-03 23:12:27 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): pb: expected conforms; pb-br: expected conforms; control-early-ack-primary: expected conforms; control-early-ack-fwd: expected conforms; silent-put: expected violation; early-ack: expected violation; early-ack: expected conforms (one operation: nothing to reorder or read stale); stale-get: expected violation; stale-get: expected conforms (one operation: nothing to reorder or read stale); sh: expected conforms; misroute: expected violation; misroute: expected conforms (one shard, or one operation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G^{expl}_{spec}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **control-early-ack-fwd**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 6 | 21 | brain03, 2026-11-02 |
| 2,3 | 20 | 20 | conforms | 0.0 | 4 | 1 h, 250 GB | 20 | 210 | brain03, 2026-11-02 |
| 2,4 | 70 | 70 | conforms | 0.2 | 4 | 1 h, 250 GB | 70 | 2485 | brain03, 2026-11-02 |
| 2,5 | 252 | 252 | conforms | 2.3 | 4 | 1 h, 250 GB | 252 | 31878 | brain03, 2026-11-02 |
| 2,6 | 924 | 924 | conforms | 26.8 | 13 | 1 h, 250 GB | 924 | 427350 | brain03, 2026-11-02 |
| 2,7 | 3432 | 3432 | conforms | 327.1 | 20 | 1 h, 250 GB | 3432 | 5891028 | brain03, 2026-11-02 |
| 3,2 | 90 | 90 | conforms | 0.3 | 2 | 1 h, 250 GB | 90 | 4095 | brain03, 2026-11-02 |
| 3,3 | 1680 | 1680 | conforms | 78.6 | 13 | 1 h, 250 GB | 1680 | 1412040 | brain03, 2026-11-02 |
| 3,4 | 34650 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,2 | 2520 | 2520 | conforms | 184.1 | 12 | 1 h, 250 GB | 2520 | 3176460 | brain03, 2026-11-02 |
| 4,3 | 369600 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,4 | 63063000 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 5,2 | 113400 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

### stateless: pair **control-early-ack-primary**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 6 | 21 | brain03, 2026-11-02 |
| 2,3 | 20 | 20 | conforms | 0.0 | 4 | 1 h, 250 GB | 20 | 210 | brain03, 2026-11-02 |
| 2,4 | 70 | 70 | conforms | 0.2 | 4 | 1 h, 250 GB | 70 | 2485 | brain03, 2026-11-02 |
| 2,5 | 252 | 252 | conforms | 1.9 | 2 | 1 h, 250 GB | 252 | 31878 | brain03, 2026-11-02 |
| 2,6 | 924 | 924 | conforms | 31.0 | 14 | 1 h, 250 GB | 924 | 427350 | brain03, 2026-11-02 |
| 2,7 | 3432 | 3432 | conforms | 394.5 | 19 | 1 h, 250 GB | 3432 | 5891028 | brain03, 2026-11-02 |
| 3,2 | 90 | 90 | conforms | 0.2 | 2 | 1 h, 250 GB | 90 | 4095 | brain03, 2026-11-02 |
| 3,3 | 1680 | 1680 | conforms | 75.8 | 12 | 1 h, 250 GB | 1680 | 1412040 | brain03, 2026-11-02 |
| 3,4 | 34650 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,2 | 2520 | 2520 | conforms | 186.7 | 12 | 1 h, 250 GB | 2520 | 3176460 | brain03, 2026-11-02 |
| 4,3 | 369600 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,4 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 5,2 | 113400 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

## stateless: pair **pb**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | 6 | 21 | brain03, 2026-11-02 |
| 2,3 | 20 | 20 | conforms | 0.0 | 2 | 1 h, 250 GB | 20 | 210 | brain03, 2026-11-02 |
| 2,4 | 70 | 70 | conforms | 0.2 | 4 | 1 h, 250 GB | 70 | 2485 | brain03, 2026-11-02 |
| 2,5 | 252 | 252 | conforms | 2.1 | 4 | 1 h, 250 GB | 252 | 31878 | brain03, 2026-11-02 |
| 2,6 | 924 | 924 | conforms | 26.3 | 18 | 1 h, 250 GB | 924 | 427350 | brain03, 2026-11-02 |
| 2,7 | 3432 | 3432 | conforms | 214.3 | 25 | 1 h, 250 GB | 3432 | 5891028 | brain03, 2026-11-02 |
| 3,2 | 90 | 90 | conforms | 0.2 | 2 | 1 h, 250 GB | 90 | 4095 | brain03, 2026-11-02 |
| 3,3 | 1680 | 1680 | conforms | 55.0 | 14 | 1 h, 250 GB | 1680 | 1412040 | brain03, 2026-11-02 |
| 3,4 | 34650 | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,2 | 2520 | 2520 | conforms | 182.2 | 12 | 1 h, 250 GB | 2520 | 3176460 | brain03, 2026-11-02 |
| 4,3 | 369600 | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,4 | 63063000 | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 5,2 | 113400 | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

## stateless: pair **pb-br**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 6 | 21 | brain03, 2026-11-02 |
| 2,3 | 20 | 20 | conforms | 0.0 | 2 | 1 h, 250 GB | 20 | 210 | brain03, 2026-11-02 |
| 2,4 | 70 | 70 | conforms | 0.2 | 4 | 1 h, 250 GB | 70 | 2485 | brain03, 2026-11-02 |
| 2,5 | 252 | 252 | conforms | 2.1 | 2 | 1 h, 250 GB | 252 | 31878 | brain03, 2026-11-02 |
| 2,6 | 924 | 924 | conforms | 26.4 | 18 | 1 h, 250 GB | 924 | 427350 | brain03, 2026-11-02 |
| 2,7 | 3432 | 3432 | conforms | 214.4 | 25 | 1 h, 250 GB | 3432 | 5891028 | brain03, 2026-11-02 |
| 3,2 | 90 | 90 | conforms | 0.2 | 4 | 1 h, 250 GB | 90 | 4095 | brain03, 2026-11-02 |
| 3,3 | 1680 | 1680 | conforms | 62.6 | 14 | 1 h, 250 GB | 1680 | 1412040 | brain03, 2026-11-02 |
| 3,4 | 34650 | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,2 | 2520 | 2520 | conforms | 109.6 | 15 | 1 h, 250 GB | 2520 | 3176460 | brain03, 2026-11-02 |
| 4,3 | 369600 | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,4 | 63063000 | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 5,2 | 113400 | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

## stateless: pair **sh**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,2,1 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | 6 | 21 | brain03, 2026-11-02 |
| 2,2,2 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | 6 | 21 | brain03, 2026-11-02 |
| 2,2,3 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | 6 | 21 | brain03, 2026-11-02 |
| 2,2,4 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | 6 | 21 | brain03, 2026-11-02 |
| 2,3,1 | 20 | 20 | conforms | 0.0 | 4 | 1 h, 250 GB | 20 | 210 | brain03, 2026-11-02 |
| 2,3,2 | 20 | 20 | conforms | 0.0 | 2 | 1 h, 250 GB | 20 | 210 | brain03, 2026-11-02 |
| 2,3,3 | 20 | 20 | conforms | 0.0 | 2 | 1 h, 250 GB | 20 | 210 | brain03, 2026-11-02 |
| 2,3,4 | 20 | 20 | conforms | 0.0 | 4 | 1 h, 250 GB | 20 | 210 | brain03, 2026-11-02 |
| 2,4,1 | 70 | 70 | conforms | 0.2 | 2 | 1 h, 250 GB | 70 | 2485 | brain03, 2026-11-02 |
| 2,4,2 | 70 | 70 | conforms | 0.2 | 2 | 1 h, 250 GB | 70 | 2485 | brain03, 2026-11-02 |
| 2,4,3 | 70 | 70 | conforms | 0.2 | 2 | 1 h, 250 GB | 70 | 2485 | brain03, 2026-11-02 |
| 2,4,4 | 70 | 70 | conforms | 0.2 | 2 | 1 h, 250 GB | 70 | 2485 | brain03, 2026-11-02 |
| 2,5,1 | 252 | 252 | conforms | 2.2 | 2 | 1 h, 250 GB | 252 | 31878 | brain03, 2026-11-02 |
| 2,5,2 | 252 | 252 | conforms | 2.0 | 2 | 1 h, 250 GB | 252 | 31878 | brain03, 2026-11-02 |
| 2,5,3 | 252 | 252 | conforms | 2.1 | 2 | 1 h, 250 GB | 252 | 31878 | brain03, 2026-11-02 |
| 2,5,4 | 252 | 252 | conforms | 2.2 | 2 | 1 h, 250 GB | 252 | 31878 | brain03, 2026-11-02 |
| 3,2,1 | 90 | 90 | conforms | 0.2 | 4 | 1 h, 250 GB | 90 | 4095 | brain03, 2026-11-02 |
| 3,2,2 | 90 | 90 | conforms | 0.2 | 4 | 1 h, 250 GB | 90 | 4095 | brain03, 2026-11-02 |
| 3,2,3 | 90 | 90 | conforms | 0.3 | 4 | 1 h, 250 GB | 90 | 4095 | brain03, 2026-11-02 |
| 3,2,4 | 90 | 90 | conforms | 0.2 | 4 | 1 h, 250 GB | 90 | 4095 | brain03, 2026-11-02 |
| 3,3,1 | 1680 | 1680 | conforms | 63.6 | 13 | 1 h, 250 GB | 1680 | 1412040 | brain03, 2026-11-02 |
| 3,3,2 | 1680 | 1680 | conforms | 63.3 | 14 | 1 h, 250 GB | 1680 | 1412040 | brain03, 2026-11-02 |
| 3,3,3 | 1680 | 1680 | conforms | 66.1 | 13 | 1 h, 250 GB | 1680 | 1412040 | brain03, 2026-11-02 |
| 3,3,4 | 1680 | 1680 | conforms | 66.2 | 13 | 1 h, 250 GB | 1680 | 1412040 | brain03, 2026-11-02 |
| 3,4,1 | 34650 | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 4,2,1 | 2520 | 2520 | conforms | 179.1 | 12 | 1 h, 250 GB | 2520 | 3176460 | brain03, 2026-11-02 |
| 4,2,2 | 2520 | 2520 | conforms | 173.2 | 12 | 1 h, 250 GB | 2520 | 3176460 | brain03, 2026-11-02 |
| 4,2,3 | 2520 | 2520 | conforms | 174.6 | 12 | 1 h, 250 GB | 2520 | 3176460 | brain03, 2026-11-02 |
| 4,2,4 | 2520 | 2520 | conforms | 172.2 | 13 | 1 h, 250 GB | 2520 | 3176460 | brain03, 2026-11-02 |
| 4,3,1 | 369600 | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 5,2,1 | 113400 | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

## stateless: pair **early-ack**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 10 | 7 | violation | 0.0 | 2 | 1 h, 250 GB | 10 | 53 | brain03, 2026-11-02 |
| 2,3 | 20 | 54 | 42 | violation | 0.1 | 4 | 1 h, 250 GB | 51 | 937 | brain03, 2026-11-02 |
| 2,4 | 70 | 300 | 269 | violation | 1.1 | 4 | 1 h, 250 GB | 293 | 19692 | brain03, 2026-11-02 |
| 2,5 | 252 | 1736 | 1588 | violation | 13.3 | 15 | 1 h, 250 GB | 1663 | 409767 | brain03, 2026-11-02 |
| 2,6 | 924 | 10102 | 9719 | violation | 182.6 | 40 | 1 h, 250 GB | 9931 | 9080757 | brain03, 2026-11-02 |
| 2,7 | 3432 | 60106 | 58212 | violation | 1896.2 | 192 | 1 h, 250 GB | 58887 | 200961061 | brain03, 2026-11-02 |
| 3,2 | 90 | 196 | 174 | violation | 0.9 | 4 | 1 h, 250 GB | 196 | 16501 | brain03, 2026-11-02 |
| 3,3 | 1680 | 5142 | 4811 | violation | 217.0 | 26 | 1 h, 250 GB | 5073 | 8262190 | brain03, 2026-11-02 |
| 3,4 | 34650 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,2 | 2520 | 4848 | 4604 | violation | 389.5 | 26 | 1 h, 250 GB | 4848 | 11866342 | brain03, 2026-11-02 |
| 4,3 | 369600 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,4 | | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 5,2 | 113400 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

## stateless: pair **misroute**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2,2,2 | 6 | 6 | 6 | violation | 0.0 | 2 | 1 h, 250 GB | 6 | 36 | brain03, 2026-11-02 |
| 2,2,3 | 6 | 6 | 6 | violation | 0.0 | 2 | 1 h, 250 GB | 6 | 36 | brain03, 2026-11-02 |
| 2,2,4 | 6 | 6 | 6 | violation | 0.0 | 2 | 1 h, 250 GB | 6 | 36 | brain03, 2026-11-02 |
| 2,3,2 | 20 | 20 | 20 | violation | 0.0 | 4 | 1 h, 250 GB | 20 | 400 | brain03, 2026-11-02 |
| 2,3,3 | 20 | 20 | 20 | violation | 0.1 | 4 | 1 h, 250 GB | 20 | 400 | brain03, 2026-11-02 |
| 2,3,4 | 20 | 20 | 20 | violation | 0.0 | 2 | 1 h, 250 GB | 20 | 400 | brain03, 2026-11-02 |
| 2,4,2 | 70 | 70 | 70 | violation | 0.3 | 2 | 1 h, 250 GB | 70 | 4900 | brain03, 2026-11-02 |
| 2,4,3 | 70 | 70 | 70 | violation | 0.3 | 2 | 1 h, 250 GB | 70 | 4900 | brain03, 2026-11-02 |
| 2,4,4 | 70 | 70 | 70 | violation | 0.3 | 2 | 1 h, 250 GB | 70 | 4900 | brain03, 2026-11-02 |
| 2,5,2 | 252 | 252 | 252 | violation | 4.2 | 4 | 1 h, 250 GB | 252 | 63504 | brain03, 2026-11-02 |
| 2,5,3 | 252 | 252 | 252 | violation | 4.2 | 2 | 1 h, 250 GB | 252 | 63504 | brain03, 2026-11-02 |
| 2,5,4 | 252 | 252 | 252 | violation | 4.2 | 4 | 1 h, 250 GB | 252 | 63504 | brain03, 2026-11-02 |
| 3,2,2 | 90 | 90 | 90 | violation | 0.5 | 4 | 1 h, 250 GB | 90 | 8100 | brain03, 2026-11-02 |
| 3,2,3 | 90 | 90 | 90 | violation | 0.5 | 2 | 1 h, 250 GB | 90 | 8100 | brain03, 2026-11-02 |
| 3,2,4 | 90 | 90 | 90 | violation | 0.5 | 4 | 1 h, 250 GB | 90 | 8100 | brain03, 2026-11-02 |
| 3,3,2 | 1680 | 1680 | 1680 | violation | 123.3 | 14 | 1 h, 250 GB | 1680 | 2822400 | brain03, 2026-11-02 |
| 3,3,3 | 1680 | 1680 | 1680 | violation | 131.4 | 14 | 1 h, 250 GB | 1680 | 2822400 | brain03, 2026-11-02 |
| 3,3,4 | 1680 | 1680 | 1680 | violation | 132.0 | 14 | 1 h, 250 GB | 1680 | 2822400 | brain03, 2026-11-02 |
| 3,4,2 | 34650 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,2,2 | 2520 | 2520 | 2520 | violation | 343.6 | 16 | 1 h, 250 GB | 2520 | 6350400 | brain03, 2026-11-02 |
| 4,2,3 | 2520 | 2520 | 2520 | violation | 341.3 | 18 | 1 h, 250 GB | 2520 | 6350400 | brain03, 2026-11-02 |
| 4,2,4 | 2520 | 2520 | 2520 | violation | 340.7 | 16 | 1 h, 250 GB | 2520 | 6350400 | brain03, 2026-11-02 |
| 4,3,2 | 369600 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 5,2,2 | 113400 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

## stateless: pair **silent-put**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 3 | 3 | violation | 0.0 | 4 | 1 h, 250 GB | 3 | 18 | brain03, 2026-11-02 |
| 2,3 | 20 | 3 | 3 | violation | 0.0 | 4 | 1 h, 250 GB | 3 | 60 | brain03, 2026-11-02 |
| 2,4 | 70 | 3 | 3 | violation | 0.0 | 2 | 1 h, 250 GB | 3 | 210 | brain03, 2026-11-02 |
| 2,5 | 252 | 3 | 3 | violation | 0.1 | 4 | 1 h, 250 GB | 3 | 756 | brain03, 2026-11-02 |
| 2,6 | 924 | 3 | 3 | violation | 0.3 | 2 | 1 h, 250 GB | 3 | 2772 | brain03, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2,7 | 3432 | 3 | 3 | violation | 1.3 | 4 | 1 h, 250 GB | 3 | 10296 | brain03, 2026-11-02 |
| 3,2 | 90 | 24 | 24 | violation | 0.1 | 2 | 1 h, 250 GB | 24 | 2160 | brain03, 2026-11-02 |
| 3,3 | 1680 | 80 | 80 | violation | 8.2 | 4 | 1 h, 250 GB | 80 | 134400 | brain03, 2026-11-02 |
| 3,4 | 34650 | 280 | 280 | violation | 647.7 | 14 | 1 h, 250 GB | 280 | 9702000 | brain03, 2026-11-02 |
| 3,5 | 756756 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,2 | 2520 | 648 | 648 | violation | 94.7 | 12 | 1 h, 250 GB | 648 | 1632960 | brain03, 2026-11-02 |
| 4,3 | 369600 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,4 | 63063000 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 5,2 | 113400 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

### stateless: pair **stale-get**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | 6 | violation | 0.0 | 4 | 1 h, 250 GB | 6 | 36 | brain03, 2026-11-02 |
| 2,3 | 20 | 20 | 20 | violation | 0.0 | 4 | 1 h, 250 GB | 20 | 400 | brain03, 2026-11-02 |
| 2,4 | 70 | 70 | 70 | violation | 0.3 | 2 | 1 h, 250 GB | 70 | 4900 | brain03, 2026-11-02 |
| 2,5 | 252 | 252 | 252 | violation | 4.1 | 2 | 1 h, 250 GB | 252 | 63504 | brain03, 2026-11-02 |
| 2,6 | 924 | 924 | 924 | violation | 48.8 | 15 | 1 h, 250 GB | 924 | 853776 | brain03, 2026-11-02 |
| 2,7 | 3432 | 3432 | 3432 | violation | 430.5 | 26 | 1 h, 250 GB | 3432 | 11778624 | brain03, 2026-11-02 |
| 3,2 | 90 | 90 | 90 | violation | 0.5 | 4 | 1 h, 250 GB | 90 | 8100 | brain03, 2026-11-02 |
| 3,3 | 1680 | 1680 | 1680 | violation | 121.1 | 14 | 1 h, 250 GB | 1680 | 2822400 | brain03, 2026-11-02 |
| 3,4 | 34650 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,2 | 2520 | 2520 | 2520 | violation | 334.0 | 14 | 1 h, 250 GB | 2520 | 6350400 | brain03, 2026-11-02 |
| 4,3 | 369600 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,4 | 63063000 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 5,2 | 113400 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): misroute 4 3 3 fifo, misroute 4 3 4 fifo, misroute 4 4 2 fifo, sh 4 4 3 fifo, misroute 4 4 3 fifo, misroute 4 4 4 fifo, misroute 4 5 2 fifo, misroute 4 5 3 fifo, misroute 4 5 4 fifo, misroute 5 2 3 fifo, misroute 5 2 4 fifo, misroute 5 3 2 fifo, misroute 5 3 3 fifo, misroute 5 3 4 fifo, misroute 5 4 2 fifo, misroute 5 4 3 fifo, misroute 5 4 4 fifo, misroute 5 5 2 fifo, misroute 5 5 3 fifo, misroute 5 5 4 fifo.

skipped: a smaller size had reached its limit: sh 3 4 2 fifo, sh 3 4 3 fifo, misroute 3 4 3 fifo, sh 3 4 4 fifo, misroute 3 4 4 fifo, stale-get 3 5 fifo, early-ack 3 5 fifo, pb-br 3 5 fifo, pb 3 5 fifo, control-early-ack-primary 3 5 fifo, control-early-ack-fwd 3 5 fifo, sh 3 5 1 fifo, sh 3 5 2 fifo, misroute 3 5 2 fifo, misroute 3 5 3 fifo, sh 3 5 3 fifo, sh 3 5 4 fifo, misroute 3 5 4 fifo, silent-put 3 6 fifo, early-ack 3 6 fifo, control-early-ack-primary 3 6 fifo, control-early-ack-fwd 3 6 fifo, pb-br 3 6 fifo, pb 3 6 fifo, … (198 instances in all).

## stateful

### stateful: pair **control-early-ack-fwd**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 20 | 20 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 70 | 70 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5 | 252 | 252 | conforms | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,6 | 924 | 924 | conforms | 0.2 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,7 | 3432 | 3432 | conforms | 0.6 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2 | 90 | 90 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 1680 | 1680 | conforms | 0.4 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 34650 | 34650 | conforms | 3.2 | 6 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5 | 756756 | 756756 | conforms | 32.2 | 247 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,6 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2 | 2520 | 2520 | conforms | 0.4 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | 369600 | 369600 | conforms | 30.2 | 114 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,4 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2 | 113400 | 113400 | conforms | 18.1 | 36 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,3 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,2 | 7484400 | 7484400 | conforms | 1580.0 | 1727 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,3 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,2 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

### stateful: pair **control-early-ack-primary**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 20 | 20 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 70 | 70 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5 | 252 | 252 | conforms | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,6 | 924 | 924 | conforms | 0.2 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,7 | 3432 | 3432 | conforms | 0.5 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2 | 90 | 90 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 1680 | 1680 | conforms | 0.4 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 34650 | 34650 | conforms | 3.0 | 6 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5 | 756756 | 756756 | conforms | 45.3 | 245 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,6 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2 | 2520 | 2520 | conforms | 0.4 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | 369600 | 369600 | conforms | 34.1 | 105 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,4 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2 | 113400 | 113400 | conforms | 15.9 | 36 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,3 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,2 | 7484400 | 7484400 | conforms | 1325.3 | 1727 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,3 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

### stateful: pair **pb**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 20 | 20 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 70 | 70 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5 | 252 | 252 | conforms | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,6 | 924 | 924 | conforms | 0.2 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,7 | 3432 | 3432 | conforms | 0.5 | 6 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2 | 90 | 90 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 1680 | 1680 | conforms | 0.3 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 34650 | 34650 | conforms | 1.8 | 16 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5 | 756756 | 756756 | conforms | 21.8 | 248 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,6 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2 | 2520 | 2520 | conforms | 0.4 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | 369600 | 369600 | conforms | 23.6 | 118 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,4 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2 | 113400 | 113400 | conforms | 14.5 | 38 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,3 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,2 | 7484400 | 7484400 | conforms | 983.4 | 1731 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,3 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,2 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

### stateful: pair **pb-br**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 20 | 20 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 70 | 70 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5 | 252 | 252 | conforms | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,6 | 924 | 924 | conforms | 0.2 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,7 | 3432 | 3432 | conforms | 0.4 | 6 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2 | 90 | 90 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 1680 | 1680 | conforms | 0.4 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 34650 | 34650 | conforms | 1.8 | 20 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5 | 756756 | 756756 | conforms | 20.8 | 250 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,6 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2 | 2520 | 2520 | conforms | 0.4 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | 369600 | 369600 | conforms | 12.0 | 121 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,4 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 5,2 | 113400 | 113400 | conforms | 8.0 | 24 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,3 | | | 🕓 (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,2 | 7484400 | 7484400 | conforms | 413.9 | 1738 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,3 | | | 🕓 (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,2 | | | 🕓 (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

## stateful: pair **sh**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2,2,1 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,2,2 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,2,3 | 6 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,2,4 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3,1 | 20 | 20 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3,2 | 20 | 20 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3,3 | 20 | 20 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3,4 | 20 | 20 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4,1 | 70 | 70 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4,2 | 70 | 70 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4,3 | 70 | 70 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4,4 | 70 | 70 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5,1 | 252 | 252 | conforms | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5,2 | 252 | 252 | conforms | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5,3 | 252 | 252 | conforms | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5,4 | 252 | 252 | conforms | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2,1 | 90 | 90 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2,2 | 90 | 90 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2,3 | 90 | 90 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2,4 | 90 | 90 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3,1 | 1680 | 1680 | conforms | 0.3 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3,2 | 1680 | 1680 | conforms | 0.3 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3,3 | 1680 | 1680 | conforms | 0.3 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3,4 | 1680 | 1680 | conforms | 0.3 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4,1 | 34650 | 34650 | conforms | 1.9 | 12 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4,2 | 34650 | 34650 | conforms | 2.0 | 16 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4,3 | 34650 | 34650 | conforms | 2.1 | 14 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4,4 | 34650 | 34650 | conforms | 2.0 | 20 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5,1 | 756756 | 756756 | conforms | 22.0 | 248 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5,2 | 756756 | 756756 | conforms | 22.5 | 249 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5,3 | 756756 | 756756 | conforms | 23.1 | 249 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5,4 | 756756 | 756756 | conforms | 23.5 | 250 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2,1 | 2520 | 2520 | conforms | 0.5 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2,2 | 2520 | 2520 | conforms | 0.6 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2,3 | 2520 | 2520 | conforms | 0.5 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2,4 | 2520 | 2520 | conforms | 0.5 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3,1 | 369600 | 369600 | conforms | 15.5 | 119 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3,2 | 369600 | 369600 | conforms | 15.7 | 120 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3,3 | 369600 | 369600 | conforms | 15.9 | 121 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3,4 | 369600 | 369600 | conforms | 15.9 | 121 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,4,1 | | | 🕓 (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2,1 | 113400 | 113400 | conforms | 15.2 | 39 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2,2 | 113400 | 113400 | conforms | 15.3 | 39 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2,3 | 113400 | 113400 | conforms | 14.9 | 40 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2,4 | 113400 | 113400 | conforms | 15.5 | 40 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,3,1 | | | 🕓 (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

## stateful: pair **early-ack**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 10 | 7 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 20 | 54 | 42 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 70 | 300 | 269 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2,5 | 252 | 1736 | 1588 | violation | 0.2 | 8 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,6 | 924 | 10102 | 9719 | violation | 0.6 | 46 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,7 | 3432 | 60106 | 58212 | violation | 2.1 | 242 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2 | 90 | 196 | 174 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 1680 | 5142 | 4811 | violation | 0.5 | 28 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 34650 | 224810 | 221544 | violation | 7.9 | 1406 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5 | 756756 | 6847694 | 6784495 | violation | 777.9 | 52912 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,6 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2 | 2520 | 4848 | 4604 | violation | 0.6 | 30 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | 369600 | 1320692 | 1294900 | violation | 54.1 | 8495 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,4 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2 | 113400 | 258768 | 254220 | violation | 14.7 | 1854 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,3 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,2 | 7484400 | 15512112 | 15403284 | violation | 1918.5 | 130648 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,3 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,2 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

### stateful: pair **misroute**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2,2,2 | 6 | 6 | 6 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,2,3 | 6 | 6 | 6 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,2,4 | 6 | 6 | 6 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3,2 | 20 | 20 | 20 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3,3 | 20 | 20 | 20 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3,4 | 20 | 20 | 20 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4,2 | 70 | 70 | 70 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4,3 | 70 | 70 | 70 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4,4 | 70 | 70 | 70 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5,2 | 252 | 252 | 252 | violation | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5,3 | 252 | 252 | 252 | violation | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5,4 | 252 | 252 | 252 | violation | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2,2 | 90 | 90 | 90 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2,3 | 90 | 90 | 90 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2,4 | 90 | 90 | 90 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3,2 | 1680 | 1680 | 1680 | violation | 0.3 | 12 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3,3 | 1680 | 1680 | 1680 | violation | 0.3 | 12 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3,4 | 1680 | 1680 | 1680 | violation | 0.3 | 12 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4,2 | 34650 | 34650 | 34650 | violation | 2.7 | 291 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4,3 | 34650 | 34650 | 34650 | violation | 2.8 | 292 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4,4 | 34650 | 34650 | 34650 | violation | 2.9 | 292 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5,2 | 756756 | 756756 | 756756 | violation | 46.0 | 8485 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5,3 | 756756 | 756756 | 756756 | violation | 46.5 | 8494 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5,4 | 756756 | 756756 | 756756 | violation | 46.6 | 8487 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2,2 | 2520 | 2520 | 2520 | violation | 0.5 | 16 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2,3 | 2520 | 2520 | 2520 | violation | 0.5 | 18 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2,4 | 2520 | 2520 | 2520 | violation | 0.6 | 18 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3,2 | 369600 | 369600 | 369600 | violation | 24.0 | 3106 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3,3 | 369600 | 369600 | 369600 | violation | 24.5 | 3102 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3,4 | 369600 | 369600 | 369600 | violation | 25.2 | 3110 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,4,2 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2,2 | 113400 | 113400 | 113400 | violation | 17.0 | 883 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2,3 | 113400 | 113400 | 113400 | violation | 18.0 | 887 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2,4 | 113400 | 113400 | 113400 | violation | 17.8 | 882 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,3,2 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

### stateful: pair **silent-put**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 3 | 3 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 20 | 3 | 3 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 70 | 3 | 3 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2,5 | 252 | 3 | 3 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,6 | 924 | 3 | 3 | violation | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,7 | 3432 | 3 | 3 | violation | 0.2 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2 | 90 | 24 | 24 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 1680 | 80 | 80 | violation | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 34650 | 280 | 280 | violation | 0.7 | 8 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5 | 756756 | 1008 | 1008 | violation | 9.9 | 172 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,6 | 17153136 | 3696 | 3696 | violation | 2034.2 | 2893 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,7 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2 | 2520 | 648 | 648 | violation | 0.3 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | 369600 | 12040 | 12040 | violation | 7.5 | 163 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,4 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2 | 113400 | 19980 | 19980 | violation | 7.8 | 168 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,2 | 7484400 | 1360800 | 1360800 | violation | 392.6 | 12853 | 1 h, 250 GB | brain02, 2026-11-02 |

### stateful: pair **stale-get**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | 6 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 20 | 20 | 20 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 70 | 70 | 70 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5 | 252 | 252 | 252 | violation | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,6 | 924 | 924 | 924 | violation | 0.2 | 6 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,7 | 3432 | 3432 | 3432 | violation | 0.5 | 34 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2 | 90 | 90 | 90 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 1680 | 1680 | 1680 | violation | 0.4 | 14 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 34650 | 34650 | 34650 | violation | 2.6 | 289 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5 | 756756 | 756756 | 756756 | violation | 45.2 | 8492 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2 | 2520 | 2520 | 2520 | violation | 0.5 | 16 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | 369600 | 369600 | 369600 | violation | 32.5 | 3112 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,4 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2 | 113400 | 113400 | 113400 | violation | 16.6 | 879 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,3 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,2 | 7484400 | 7484400 | 7484400 | violation | 1306.8 | 66779 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6,3 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 7,2 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

**Not run (stateful).** planned but never scheduled (grid points beyond the ladder): stale-get 3 6 fifo, silent-put 5 3 fifo, pb 6 2 fifo, silent-put 6 3 fifo, misroute 6 6 4 fifo, sh 6 6 4 fifo, control-early-ack-primary 7 2 fifo, silent-put 7 2 fifo, control-early-ack-primary 7 3 fifo, control-early-ack-fwd 7 3 fifo, control-early-ack-fwd 7 4 fifo, control-early-ack-fwd 7 5 fifo, control-early-ack-fwd 7 6 fifo, control-early-ack-fwd 7 7 fifo.

skipped: a smaller size had reached its limit: stale-get 3 7 fifo, control-early-ack-fwd 3 7 fifo, control-early-ack-primary 3 7 fifo, pb-br 3 7 fifo, pb 3 7 fifo, early-ack 3 7 fifo, sh 4 4 2 fifo, sh 4 4 3 fifo, misroute 4 4 3 fifo, misroute 4 4 4 fifo, sh 4 4 4 fifo, control-early-ack-primary 4 5 fifo, pb 4 5 fifo, silent-put 4 5 fifo, pb-br 4 5 fifo, stale-get 4 5 fifo, early-ack 4 5 fifo, control-early-ack-fwd 4 5 fifo, sh 4 5 1 fifo, misroute 4 5 2 fifo, sh 4 5 2 fifo, misroute 4 5 3 fifo, sh 4 5 3 fifo, sh 4 5 4 fifo, … (143 instances in all).

## offline

### offline: pair **control-early-ack-fwd**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 1680 | 1680 | conforms | 0.6 | 20 | 10 min, 16 GB | 0.1 | 0.4 | 0.1 | in-memory | brain01, 2026-11-02 |
| 4,4 | 63063000 | 63063000 | oom (250 GB) | | | 1 h, 250 GB | 1606.5 | 2847.2 | | in-memory | brain01, 2026-11-02 |

### offline: pair **control-early-ack-primary**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 1680 | 1680 | conforms | 0.3 | 20 | 10 min, 16 GB | 0.1 | 0.2 | 0.1 | in-memory | brain01, 2026-11-02 |

### offline: pair **pb**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 1680 | 1680 | conforms | 0.4 | 20 | 10 min, 16 GB | 0.1 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 4,4 | 63063000 | 63063000 | (1 h) | | | 1 h, 250 GB | 1606.5 | 1980.4 | | merge | brain01, 2026-11-03 |

### offline: pair **pb-br**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.3 | 4 | 10 min, 16 GB | 0.1 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 1680 | 1680 | conforms | 0.4 | 20 | 10 min, 16 GB | 0.1 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 4,4 | 63063000 | 63063000 | oom (250 GB) | | | 1 h, 250 GB | 1606.5 | 2001.2 | | in-memory | brain01, 2026-11-02 |

### offline: pair **sh**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2,1 | 6 | 6 | conforms | 0.3 | 4 | 10 min, 16 GB | 0.2 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2,2,2 | 6 | 6 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.2 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3,1 | 1680 | 1680 | conforms | 0.4 | 20 | 10 min, 16 GB | 0.3 | 0.2 | 0.1 | in-memory | brain01, 2026-11-02 |
| 3,3,2 | 1680 | 1680 | conforms | 0.5 | 20 | 10 min, 16 GB | 0.4 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |

### offline: pair **early-ack**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 10 | 7 | violation | 0.3 | 4 | 10 min, 16 GB | 0.1 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 1680 | 5142 | 4094 | violation | 0.9 | 38 | 10 min, 16 GB | 0.1 | 0.7 | 0.2 | in-memory | brain01, 2026-11-02 |

### offline: pair **misroute**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2,2 | 6 | 6 | 6 | violation | 0.2 | 4 | 10 min, 16 GB | 0.2 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3,2 | 1680 | 1680 | 1680 | violation | 0.5 | 22 | 10 min, 16 GB | 0.4 | 0.2 | 0.1 | in-memory | brain01, 2026-11-02 |

### offline: pair **silent-put**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 3 | 3 | violation | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 1680 | 80 | 80 | violation | 0.2 | 12 | 10 min, 16 GB | 0.1 | 0.0 | 0.1 | in-memory | brain01, 2026-11-02 |
| 4,4 | 63063000 | 247800 | | oom (250 GB) | | | 1 h, 250 GB | 1606.5 | 12.8 | | in-memory | brain01, 2026-11-02 |

### offline: pair **stale-get**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | 6 | violation | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 1680 | 1680 | 1680 | violation | 0.3 | 22 | 10 min, 16 GB | 0.1 | 0.2 | 0.1 | in-memory | brain01, 2026-11-02 |
| 4,4 | 63063000 | 63063000 | | oom (250 GB) | | | 1 h, 250 GB | 1606.5 | 1945.6 | | in-memory | brain01, 2026-11-02 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): sh 2 2 3 fifo, misroute 2 2 3 fifo, sh 2 2 4 fifo, misroute 2 2 4 fifo, stale-get 2 3 fifo, pb 2 3 fifo, control-early-ack-fwd 2 3 fifo, early-ack 2 3 fifo, control-early-ack-primary 2 3 fifo, pb-br 2 3 fifo, silent-put 2 3 fifo, sh 2 3 1 fifo, misroute 2 3 2 fifo, sh 2 3 2 fifo, misroute 2 3 3 fifo, sh 2 3 3 fifo, sh 2 3 4 fifo, misroute 2 3 4 fifo, stale-get 2 4 fifo, pb-br 2 4 fifo, control-early-ack-fwd 2 4 fifo, silent-put 2 4 fifo, early-ack 2 4 fifo, pb 2 4 fifo, ... (342 instances in all).

skipped: a smaller size had reached its limit: control-early-ack-primary 4 4 fifo, early-ack 4 4 fifo.

# context-free twins offline, Rust implementation against the context-free Java port

## context-free twins offline, Rust implementation against the context-free Java port: pair control-early-ack-fwd, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 4.4 | 162 | 10 min, 16 GB | 4.3 | 0.2 | 0.1 | in-memory | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 13.4 | 5076 | 10 min, 16 GB | 13.3 | 0.4 | 0.1 | in-memory | brain02, 2026-11-03 |

## context-free twins offline, Rust implementation against the context-free Java port: pair control-early-ack-primary, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 3.9 | 153 | 10 min, 16 GB | 3.8 | 0.2 | 0.1 | in-memory | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 10.8 | 4504 | 10 min, 16 GB | 10.7 | 0.2 | 0.1 | in-memory | brain02, 2026-11-03 |

## context-free twins offline, Rust implementation against the context-free Java port: pair early-ack, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 10 | 10 | conforms | 4.6 | 167 | 10 min, 250 GB | 4.6 | 0.3 | 0.0 | merge | brain02, 2026-11-03 |
| 3,3 | 5142 | 5142 | conforms | 20.7 | 3046 | 10 min, 250 GB | 20.4 | 0.7 | 0.3 | merge | brain02, 2026-11-03 |

## context-free twins offline, Rust implementation against the context-free Java port: pair misroute, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2,2 | 6 | 6 | conforms | 4.1 | 158 | 10 min, 16 GB | 4.1 | 0.2 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,3,2 | 1680 | 1680 | conforms | 19.0 | 4369 | 10 min, 16 GB | 18.9 | 0.9 | 0.1 | in-memory | brain02, 2026-11-03 |

## context-free twins offline, Rust implementation against the context-free Java port: pair pb, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 4.5 | 152 | 10 min, 250 GB | 4.5 | 0.3 | 0.0 | merge | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 12.9 | 2617 | 10 min, 250 GB | 12.7 | 0.3 | 0.2 | merge | brain02, 2026-11-03 |

## context-free twins offline, Rust implementation against the context-free Java port: pair pb-br, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 3.4 | 156 | 10 min, 16 GB | 3.4 | 0.0 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 11.4 | 4250 | 10 min, 16 GB | 11.3 | 0.3 | 0.1 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **sh**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2,1 | 6 | 6 | conforms | 6.1 | 166 | 10 min, 16 GB | 6.1 | 0.2 | 0.0 | in-memory | brain02, 2026-11-03 |
| 2,2,2 | 6 | 6 | conforms | 4.1 | 153 | 10 min, 16 GB | 4.1 | 0.2 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,3,1 | 1680 | 1680 | conforms | 12.8 | 4715 | 10 min, 16 GB | 12.7 | 0.3 | 0.1 | in-memory | brain02, 2026-11-03 |
| 3,3,2 | 1680 | 1680 | conforms | 12.8 | 4418 | 10 min, 16 GB | 12.7 | 0.3 | 0.1 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **silent-put**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 3 | 3 | conforms | 3.6 | 147 | 10 min, 16 GB | 3.6 | 0.2 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,3 | 80 | 80 | conforms | 4.5 | 156 | 10 min, 16 GB | 4.5 | 0.2 | 0.0 | in-memory | brain02, 2026-11-03 |
| 4,4 | 247800 | 247800 | conforms | 1246.4 | 20150 | 10 min, 16 GB | 1215.0 | 13.5 | 31.4 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **stale-get**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 3.8 | 157 | 10 min, 16 GB | 3.8 | 0.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 10.5 | 2668 | 10 min, 16 GB | 10.4 | 0.3 | 0.1 | in-memory | brain02, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** skipped: a smaller size had reached its limit: pb 4 4 fifo, control-early-ack-primary 4 4 fifo, early-ack 4 4 fifo.
refused (an input missing at run time): stale-get 4 4 fifo, pb-br 4 4 fifo, control-early-ack-fwd 4 4 fifo.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **control-early-ack-fwd**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 4.4 | 162 | 10 min, 16 GB | 0.2 | 4.3 | 0.1 | in-memory | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 13.4 | 5076 | 10 min, 16 GB | 0.4 | 13.3 | 0.1 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **control-early-ack-primary**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 3.9 | 153 | 10 min, 16 GB | 0.2 | 3.8 | 0.1 | in-memory | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 10.8 | 4504 | 10 min, 16 GB | 0.2 | 10.7 | 0.1 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **early-ack**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 10 | 10 | conforms | 4.6 | 167 | 10 min, 250 GB | 0.3 | 4.6 | 0.0 | merge | brain02, 2026-11-03 |
| 3,3 | 5142 | 5142 | conforms | 20.7 | 3046 | 10 min, 250 GB | 0.7 | 20.4 | 0.3 | merge | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **misroute**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2,2 | 6 | 6 | conforms | 4.1 | 158 | 10 min, 16 GB | 0.2 | 4.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,3,2 | 1680 | 1680 | conforms | 19.0 | 4369 | 10 min, 16 GB | 0.9 | 18.9 | 0.1 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **pb**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 4.5 | 152 | 10 min, 250 GB | 0.3 | 4.5 | 0.0 | merge | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 12.9 | 2617 | 10 min, 250 GB | 0.3 | 12.7 | 0.1 | merge | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **pb-br**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 3.4 | 156 | 10 min, 16 GB | 0.0 | 3.4 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 11.4 | 4250 | 10 min, 16 GB | 0.3 | 11.3 | 0.1 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **sh**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2,1 | 6 | 6 | conforms | 6.1 | 166 | 10 min, 16 GB | 0.2 | 6.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 2,2,2 | 6 | 6 | conforms | 4.1 | 153 | 10 min, 16 GB | 0.2 | 4.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,3,1 | 1680 | 1680 | conforms | 12.8 | 4715 | 10 min, 16 GB | 0.3 | 12.7 | 0.1 | in-memory | brain02, 2026-11-03 |
| 3,3,2 | 1680 | 1680 | conforms | 12.8 | 4418 | 10 min, 16 GB | 0.3 | 12.7 | 0.1 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **silent-put**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 3 | 3 | conforms | 3.6 | 147 | 10 min, 16 GB | 0.2 | 3.6 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,3 | 80 | 80 | conforms | 4.5 | 156 | 10 min, 16 GB | 0.2 | 4.5 | 0.0 | in-memory | brain02, 2026-11-03 |
| 4,4 | 247800 | 247800 | conforms | 1246.5 | 20150 | 10 min, 16 GB | 13.5 | 1215.0 | 31.5 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **stale-get**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 3.8 | 157 | 10 min, 16 GB | 0.1 | 3.8 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 10.5 | 2668 | 10 min, 16 GB | 0.3 | 10.4 | 0.1 | in-memory | brain02, 2026-11-03 |

**Not run (context-free twins offline, Java port against the Rust implementation).** skipped: a smaller size had reached its limit: control-early-ack-primary 4 4 fifo, pb 4 4 fifo, control-early-ack-fwd 4 4 fifo, pb-br 4 4 fifo, stale-get 4 4 fifo, early-ack 4 4 fifo.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **control-early-ack-fwd**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 3.6 | 174 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 13.2 | 4868 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4,4 | 63063000 | | oom (250 GB) | | | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **control-early-ack-primary**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 3.9 | 170 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 12.2 | 4723 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **early-ack**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 10 | 10 | conforms | 4.8 | 154 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 5142 | 5142 | conforms | 26.8 | 4882 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **misroute**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2,2 | 6 | 6 | conforms | 3.4 | 164 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3,2 | 1680 | 1680 | conforms | 11.7 | 4832 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **pb**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 3.9 | 160 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 14.1 | 3106 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **pb-br**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 3.4 | 158 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 13.3 | 4704 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4,4 | 63063000 | | oom (250 GB) | | | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **sh**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2,1 | 6 | 6 | conforms | 3.6 | 166 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2,2 | 6 | 6 | conforms | 3.4 | 159 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3,1 | 1680 | 1680 | conforms | 12.5 | 4762 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3,2 | 1680 | 1680 | conforms | 12.3 | 4889 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **silent-put**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 3 | 3 | conforms | 3.7 | 158 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 80 | 80 | conforms | 5.0 | 180 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4,4 | 247800 | 247800 | conforms | 1211.5 | 58383 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **stale-get**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 3.3 | 160 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 11.3 | 4883 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4,4 | 63063000 | | oom (250 GB) | | | 1 h, 250 GB | | brain02, 2026-11-03 |

**Not run (context-free twins online, Java port (JMC) against the Rust implementation's export).** skipped: a smaller size had reached its limit: pb 4 4 fifo, control-early-ack-primary 4 4 fifo, early-ack 4 4 fifo.

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **control-early-ack-fwd**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 0.4 | 14 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **control-early-ack-primary**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 0.3 | 12 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **early-ack**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 10 | 10 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 5142 | 5142 | conforms | 0.6 | 32 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **misroute**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2,2 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3,2 | 1680 | 1680 | conforms | 0.3 | 14 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **pb**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.1 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 0.3 | 14 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **pb-br**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 0.3 | 14 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **sh**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2,1 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2,2,2 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3,1 | 1680 | 1680 | conforms | 0.3 | 14 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3,2 | 1680 | 1680 | conforms | 0.4 | 14 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **silent-put**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 3 | 3 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 80 | 80 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4,4 | 247800 | 247800 | conforms | 31.8 | 2646 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **stale-get**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 1680 | 1680 | conforms | 0.3 | 14 | 1 h, 250 GB | | brain02, 2026-11-03 |

**Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).** skipped: a smaller size had reached its limit: control-early-ack-primary 4 4 fifo, pb 4 4 fifo, control-early-ack-fwd 4 4 fifo, pb-br 4 4 fifo, stale-get 4 4 fifo, early-ack 4 4 fifo.